

Multiple outcomes and analyses in clinical trials create challenges for interpretation and research synthesis

Evan Mayo-Wilson^{a,*}, Nicole Fusco^a, Tianjing Li^a, Hwanhee Hong^b, Joseph K. Canner^c, Kay Dickersin^a, for the MUDS investigators

^aDepartment of Epidemiology, Johns Hopkins University Bloomberg School of Public Health, 615 North Wolfe Street, Baltimore, MD 21205, USA

^bDepartment of Mental Health, Johns Hopkins University Bloomberg School of Public Health, 624 N Broadway, Hampton House, Baltimore, MD 21205, USA

^cDepartment of Surgery, Johns Hopkins University School of Medicine, 600 North Wolfe Street, Blalock Building, Baltimore, MD 21287, USA

Accepted 9 May 2017; Published online 18 May 2017

Abstract

Objective: To identify variations in outcomes and results across reports of randomized clinical trials (RCTs).

Study Design and Setting: Eligible RCTs examined gabapentin for neuropathic pain and quetiapine for bipolar depression, reported in public (e.g., journal articles) and nonpublic (e.g., clinical study reports) sources by 2015. We prespecified outcome domains. From each source, we collected “outcomes” (i.e., domain, measure, metric, method of aggregation, and time point); “treatment effect” (i.e., outcome plus the methods of analysis [e.g., how missing data were handled]); and results (i.e., numerical contrasts of treatment and comparison groups). We assessed whether results included sufficient information for meta-analysis.

Results: We found 21 gabapentin (68 public, 6 nonpublic reports) and seven quetiapine RCTs (46 public, 4 nonpublic reports). For four (gabapentin) and seven (quetiapine) prespecified outcome domains, RCTs reported 214 and 81 outcomes by varying four elements. RCTs assessed 605 and 188 treatment effects by varying the analysis of those outcomes. RCTs reported 1,230 and 661 meta-analyzable results, 305 (25%) and 109 (16%) in public reports.

Conclusion: RCTs included hundreds of outcomes and results; a small proportion were in public reports. Trialists and meta-analysts may cherry-pick what they report from multiple sources of RCT information. © 2017 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Keywords: Clinical trials; Systematic reviews; Meta-analysis; Outcomes; Selective outcome reporting

1. Introduction

Although randomized clinical trials (RCTs) are considered the reference standard for examining effectiveness and safety of treatments, it is rare that a single RCT provides sufficient evidence to merit adoption of a treatment for any given condition. Furthermore, clinicians and others can no

longer stay abreast of rapidly growing knowledge, including the findings of all RCTs pertinent to their treatment decisions. Accordingly, they look to summaries of knowledge, such as clinical practice guidelines, that depend in part on evidence syntheses (e.g., systematic reviews, meta-analyses); evidence syntheses combine information from similar studies, often focusing on RCTs for treatment decisions.

In many systematic reviews, “outcomes” are not well defined. Although “outcomes” are often described by a “name” such as “pain intensity,” this name is actually the “outcome domain,” one of five elements comprising an outcome [1]. The five elements are as follows: (1) outcome domain; (2) measure (e.g., McGill Pain Questionnaire, Montgomery Åsberg Depression Rating Scale); (3) metric (e.g., value at a time point, change from baseline); (4) method of aggregation (e.g., mean value for continuous data, percent with an outcome for categorical data); and (5) time point at which the assessment was made (e.g., 8 weeks after starting treatment). Thus, for a single outcome domain, one RCT may include many defined outcomes

The MUDS investigators include Lorenzo Bertizzolo, Terrie Cowley, Peter Doshi, Jeffrey Ehmsen, Gillian Gresham, Nan Guo, Jennifer Haythornthwaite, James Heyward, Diana Lock, Jennifer Payne, Lori Rosman, Elizabeth Stuart, Catalina Suarez-Cuervo, Elizabeth Tolbert, Claire Twose, and Swaroop Vedula.

Ethics approval: The study received an exemption from the Johns Hopkins Bloomberg School of Public Health Institutional Review Board (IRB no: 00006324).

Funding: Supported by contract ME 1303 5785 from the Patient-Centered Outcomes Research Institute (PCORI) and a fund established at Johns Hopkins for scholarly research on reporting biases by Greene LLP. The funders were not involved in the design or conduct of the study, article preparation, or the decision to submit the article for publication.

* Corresponding author. Tel.: 443-287-5042.

E-mail address: evan.mayo-wilson@jhu.edu (E. Mayo-Wilson).

What is new?

Key findings

- Trials of the same intervention and condition included hundreds of different outcomes and results, and much of this information was available only using nonpublic sources.

What this adds to what was known?

- Multiple outcome definitions and multiple methods of analysis lead to challenges for interpreting clinical trials, particularly because they create opportunities for cherry-picking by both clinical trialists and systematic reviewers.
- Variation in outcomes, and incomplete results reporting, makes it difficult to compare clinical trials and to translate knowledge into practice.

What is the implication and what should change now?

- Clinical trials and systematic reviews should define their outcomes and methods of analysis completely and report their results transparently.
- Guidance is needed for using multiple outcomes and results in systematic reviews.

because different measures, metrics, time points, and methods of aggregation were used (Fig. 1).

Investigators performing evidence syntheses usually prespecify eligibility criteria for including RCTs and outcome domains that will be examined. It is not unusual for investigators to find, however, that even when many trials are eligible for a systematic review, only a few trials can be combined using meta-analysis [2,3]. Consequently, many trials that are eligible for systematic reviews are not included in the meta-analyses they contain; those trials thus contribute little information to the overall conclusions of systematic reviews. This may occur because the included RCTs did not assess the same outcome domains because different outcomes within the same domains could not be combined in meta-analysis (i.e., one or more elements differed), because the included RCTs assessed but did not report the same outcomes, or because RCTs reported the same outcomes but did not report sufficient statistical information to allow combination of the numerical results [4,5]. Furthermore, if systematic reviewers assume that outcomes within RCTs can be used interchangeably, even if those outcomes are not defined using all elements, reviewers may be making assumptions that lead to errors when synthesizing overall results [6,7].

The fact that RCTs may assess multiple outcomes for the same domain leads to challenges for systematic reviewers, regardless of whether they conduct meta-

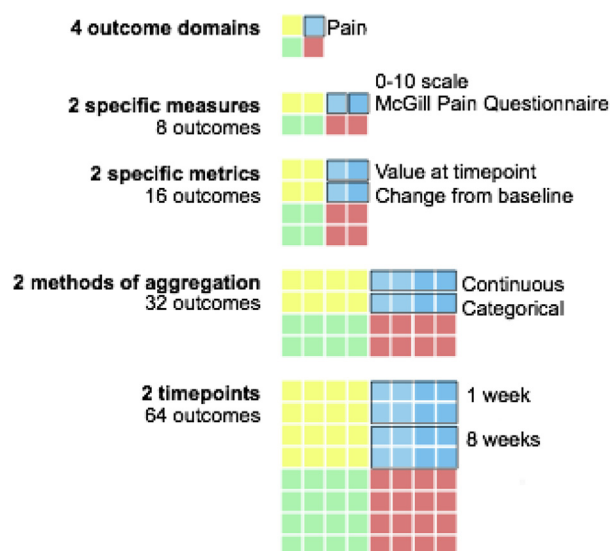


Fig. 1. The number of outcomes in a trial is a function of the number of definitions of each of the five elements. In this hypothetical example, the number of treatment effects for a single trial is the product of definitions for each element. In a trial with four outcome domains, introducing two definitions for each of the other elements will result in 64 unique defined outcomes.

analyses [8,9]. First, if an RCT reports multiple outcomes, which outcome should be used to determine whether the intervention “works”? Second, a single RCT might report different results for the same outcome by using multiple methods of analyses (e.g., methods for handling missing data) [10–14]. If there are multiple results for an outcome, which estimate should the meta-analyst use? Third, even when it is possible to combine multiple RCTs, synthesized results (e.g., the combined standardized mean difference [SMD]) may be difficult to interpret if studies used different outcome definitions or different methods of analysis [12]. All of these situations pose challenges to the proper interpretation of RCTs and evidence syntheses, and they may lead to innocent errors.

Defining multiple outcomes under the same domain may also be associated with deliberate efforts (e.g., by trialists or systematic reviewers) to conceal findings and to mislead readers. For example, in RCTs that include many outcomes, trialists might report statistically significant results selectively [14–16]. In systematic reviews, investigators might cherry-pick results to include in meta-analyses [17,18]. Furthermore, when only some outcomes are reported publicly, it is impossible for the systematic reviewer or other interpreter of the trial findings to know for sure whether there has been selective reporting.

Few studies have explored the number of results that investigators could select to include in meta-analyses [7,13,19]. We know of no studies that have used both public and nonpublic data sources for RCTs to quantify the number of outcomes and results reported across RCTs, the number of reported outcomes that are defined, or the number of results that are meta-analyzable.

2. Objective

Starting with prespecified outcome domains, we sought to identify all defined outcomes and treatment effects available from reports of eligible RCTs. We also sought to determine whether results for those outcomes were meta-analyzable and whether the information was publicly available.

3. Methods

This study was part of an investigation into multiple reports of RCTs and the effect of disagreement among reports of RCTs. Inclusion criteria, methods for the study, and results of the searches have been described elsewhere [20,21]. Definitions of terms are provided in the text and summarized in [Box 1](#).

Briefly, we identified multiple reports emanating from RCTs of (1) gabapentin treatment for neuropathic pain and (2) quetiapine treatment for bipolar depression. We did not include individual participant data (IPD) in this sub-study because information required for our analysis is not usually specified in IPD (e.g., information about metrics and methods of analysis might be included in study protocols rather than the IPD data sets). We identified public (e.g., journal articles, Food and Drug Administration [FDA] reviews, short reports) and nonpublic reports (clinical study reports [CSRs], CSR-synopses) available for these trials by 2015. We searched electronic databases for published and unpublished reports, searched for trial registrations, requested reports and data sets from manufacturers, searched the FDA web site, and checked reference lists of systematic reviews. For gabapentin-neuropathic pain trials, we hand-searched conference proceedings to identify abstracts. For quetiapine, we did not conduct hand

searches because of limitations in funding. Two authors extracted data independently from each report using the open access Systematic Review Data Repository [23] and reconciled differences through discussion.

From each eligible report, we identified outcomes and methods of analysis related to outcome domains specified a priori. All five prespecified outcome domains for gabapentin-neuropathic pain related to the potential effectiveness of treatment; two of the eight outcome domains for quetiapine-bipolar depression related to potential safety (i.e., suicide and weight).

3.1. Identifying outcomes in trial reports

We defined each “outcome” using the five elements described in the Section 1 and [Table 1](#) [1]: the (1) outcome domain, (2) measure, (3) metric, (4) method of aggregation; and (5) time point (e.g., 8 weeks after starting treatment).

When all five elements were defined, we describe an outcome as “defined”; when some but not all elements of an outcome were defined, we refer to an outcome as “not defined” ([Supplement A](#) at www.jclinepi.com).

3.2. Identifying treatment effects in trial reports

We use the term “treatment effect” to describe the outcome (when all five elements of an outcome are described the outcome is “defined”) plus the “method of analysis.” We use the term “method of analysis” to include the elements: “analysis population,” “method for handling missing data,” and “method of adjustment” ([Table 1](#)) [26,27]. We use the term “analysis population” to describe participants eligible for analysis [28]. For example, the effect of a drug might be assessed for all randomized participants or for people who took a minimum dose of the

Box 1. Definitions used in this study for key terms

Term	Definition used in our study
Outcome	An event in a person, used to assess a treatment's effect [22]. May be defined using all elements or not defined.
Defined outcome	Includes all five elements of an outcome: (1) outcome domain, (2) specific measure, (3) specific metric, (4) method of aggregation, and (5) time point. For example, “proportion of participants with 50% change from baseline to 8 weeks on the MADRS total score.”
Not defined outcome	Includes the name of an outcome domain but does not include one or more of the other four elements; for example, “symptoms of depression at 8 weeks.”
Defined treatment effect	A defined outcome plus all three methods of analysis (i.e., the analysis population, method for handling missing data, and method of adjustment).
Not defined treatment effect	Includes the name of an outcome domain but does not include one or more elements of an outcome or methods of analysis.
Result	A numerical contrast between a treatment and comparison group (e.g., relative risk, mean difference).
Meta-analyzable results	A result for which sufficient information was provided to calculate the between group effect (e.g., a point estimate and a measure of precision).
Unique (outcome, treatment effect, or result)	Defined outcome, treatment effect, or result counted only once, regardless of how many times it appeared in all reports.
Nonunique (outcome, treatment effect, or result)	Defined outcome, treatment effect, or result counted each time it appears in reports.

Abbreviation: MADRS, Montgomery Åsberg Depression Rating Scale.

Table 1. Information needed to define a treatment effect

Element	Definition used in our study	Example
A. Outcome (adapted from Zarin et al. 2011) [11]		
1. Domain	Title [24] or concept [25] to describe one or more outcomes.	Depression
2. Specific measure	Instrument used to assess the outcome domain, including the following: a. Name of the instrument or questionnaire.	Montgomery Åsberg Depression Rating Scale (MADRS).
3. Specific metric	b. The total score or the subscales that will be analyzed. Unit of measurement (e.g., value at a time point, change from baseline, time-to-event).	MADRS total score. Change from baseline.
4. Method of aggregation	The procedure for estimating the treatment effect, including the following: a. If the outcome will be treated as a continuous, categorical, or time-to-event variable.	Categorical variable.
	b. For continuous variables, a measure of central tendency (e.g., mean value). For categorical and time-to-event data variables, proportion with an event and, if relevant, the specific cutoff or categories compared.	Proportion of participants with $\geq 50\%$ decrease.
5. Time point	The period of follow-up, including the following: a. When outcome measurements will be obtained b. Which of the outcome measurements will be analyzed	Analyzed at 8 weeks after starting treatment
B. Method of analysis		
6. Analysis population	Participants eligible to be included in the analysis.	All participants taking at least one dose of study medication or placebo.
7. Method for handling missing data	a. Procedure(s) used to account for participants who withdrew from the trial, did not complete an assessment, or otherwise did not provide data. b. Procedures used for missing data items (e.g., questions that were not completed on a questionnaire). ^a	a. Last observation carried forward. b. If at least 80% of the questionnaire is complete at 8 weeks, the average score will be used for the missing item(s). If the questionnaire is less than 80% complete, the questionnaire will be treated as missing.
8. Method of adjustment	Specific analytic procedures used, if any, including the following: a. Methods to adjust the data for covariates. b. Methods to transform the data (e.g., log transformed). ^b	a. Not adjusted. b. Not transformed.

To conduct an analysis, each element of an outcome must be defined. In addition, the analyst must select a method of analysis. The right column illustrates the definition of an outcome and additional elements to define the treatment effect. This hypothetical example of a defined treatment effect in the “Example” column might be described as follows: “Depression was measured using the Montgomery Åsberg Depression Rating Scale (MADRS) total score. We compared the proportion of participants with $\geq 50\%$ decrease between baseline and 8 weeks after starting treatment, adjusting for baseline covariates. If a participant completed at least 80% of the MADRS, we used the average score; if the MADRS was less than 80% complete, we treated it as missing. All participants taking at least one dose of study medication or placebo were included in the analysis, and we imputed missing cases using last observation carried forward (LOCF).

^a Methods for handling missing items are often not described in journal articles and other public reports, and we did not record this information in this study.

^b We assumed that data were not transformed or adjusted unless otherwise specified. Thus, we recorded that this element was always defined for the purpose of our study.

assigned treatment, two different analysis populations. We did not count subgroup analyses (e.g., men vs. women) as separate analysis populations. We refer to “method for handling missing data” as the manner in which data for participants who missed an assessment or dropped out of the trial completely were considered in the analysis. We used the term “method of adjustment” to describe whether each analysis was adjusted or unadjusted for covariates and the method used. We assumed that the analysis was unadjusted unless otherwise specified, and when the analysis was adjusted did not record which specific covariates were used for adjustment [29].

3.3. Identifying results in trial reports

We use the term “result” to refer to estimates of treatment effects; that is, numerical contrasts between the test intervention and comparison groups (e.g., relative risk, mean difference) (Box 1). We extracted results for defined outcomes and for not defined outcomes (see Box 1 for definitions). We refer to “meta-analyzable results” as those results for which sufficient information was provided to include them in a mathematical synthesis with other studies (e.g., a point estimate and a measure of precision for continuous results, numerator and denominator for

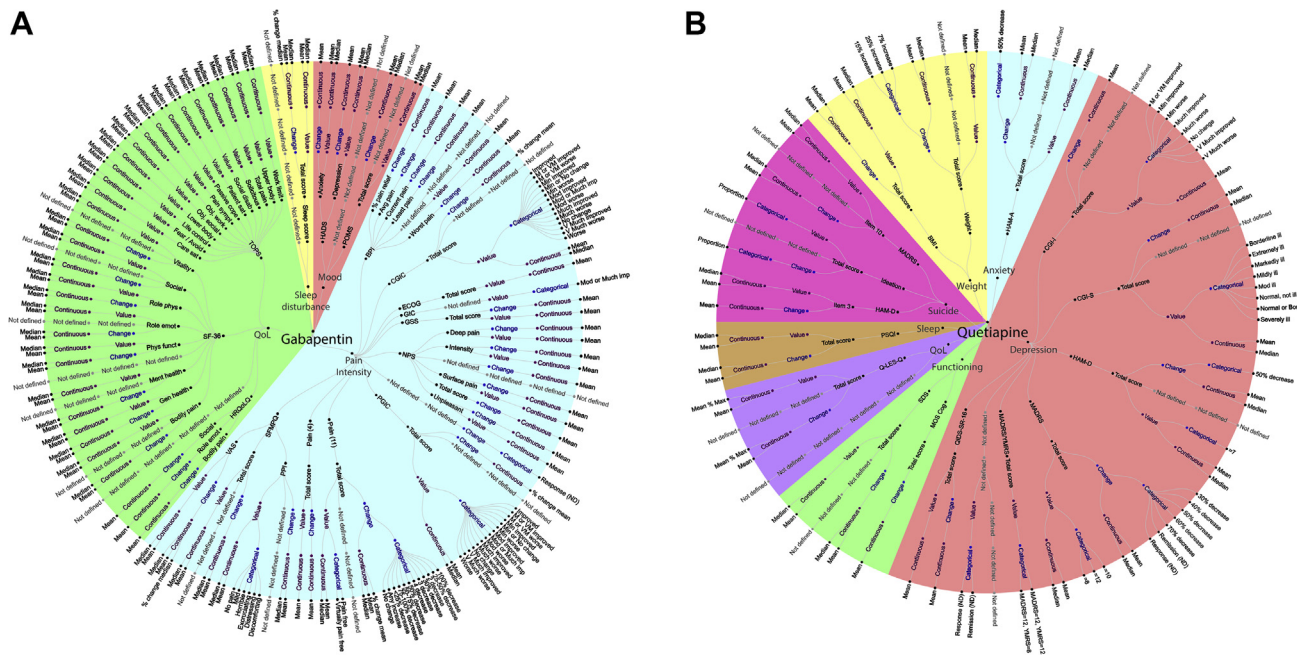


Fig. 2. The number of outcomes observed increased as a consequence of multiple definitions of the elements. The points nearest to the center of the circle represent the prespecified outcome domains for which we found any information in eligible trials (i.e., four outcome domains for gabapentin and seven outcome domains for quetiapine); these are represented by different colors. The branches represent variations in outcome definitions associated with those domains (i.e., different measures, metrics, methods of aggregation, or time points). Each point around the outer edge of the circle thus represents a unique outcome described in one or more of the trial reports we identified. (A) Gabapentin for neuropathic pain. Specific measure (name of instrument): BPI, Brief Pain Inventory; CGIC, Clinician Global Impression of Change; ECOG, Eastern Cooperative Oncology Group; GIC, Global Impression of Change; GSS, Global Symptom Score; HADS, Hospital Anxiety and Depression Scale; HRQoLQ, Health Related Quality of Life Questionnaire; NPS, Neuropathic Pain Scale; Pain (11 pt), pain score (11-point scale); Pain (4 pt), pain score (4-point scale); PGIC, Patient Global Impression of Change; POMS, Profile of Mood States; SF-36, Short Form 36 item; SFMPQ, Short Form McGill Pain Questionnaire; Sleep score, sleep score (11-point scale); TOPS, Treatment Outcomes in Pain Survey. Specific measure (total or subscale): % pain relief, percentage pain relief; Ave. pain, average pain score; Care sat, health care satisfaction; Current pain, current pain score; Fear/Avoid, fear avoidance; Gen health, general health; Globe intensity, global intensity; Globe unpleas, global unpleasantness; Lower body, lower body functional limitations; Ment health, mental health; Obj. social, objective family/social disability; Obj. work, objective work disability; Pain sympt, pain symptom; Passive cope, passive coping; Patient sat, patient satisfaction with outcomes; Phys funct, physical functioning; PPI, present pain intensity; Role emot, role emotional; Role phys, role physical; Social disab, perceived family/social disability; Social funct, social functioning; Solicitous, solicitous responses; Total pain, total pain experience; Total score, total; Upper body, upper body functional limitations; VAS, visual analog scale; Work limit, work limitations. Metric: Change, change from baseline; Value, value at a time point. Method of aggregation: % change mean, percent change mean; % change median, percent change median; 2 decrease, at least 2-point decrease; 3 decrease, at least 3-point decrease; 4 decrease, at least 4-point decrease; M or VM improved, much or very much improved; M or VM worse, much worse or very much worse; Min or No change, minimally improved or no change; Min improved, minimally improved; Min worse, minimally worse; Mod improved, moderately improved; Mod or much imp, moderately or much improved; Mod worse, moderately worse; No change, no change (0%); Pain free, totally pain free; Response (ND), response (not defined); V much improved, very much improved; V much worse, very much worse. (B) Quetiapine for bipolar depression Specific measure (name of instrument): BMI, body mass index; CGI-I, Clinical Global Impression—Improvement; CGI-S, Clinical Global Impression—Severity; HAM-A, Hamilton Anxiety Rating Scale; HAM-D, Hamilton Depression Rating Scale; MADRS, Montgomery Åsberg Depression Rating Scale; MADRS/YMRS, Montgomery Åsberg Depression Rating Scale & Young Mania Rating Scale; MOS Cog, Medical Outcomes Study Cognitive Scale; PSQI, Pittsburgh Sleep Quality Assessment; Q-LES-Q, Quality of Life Enjoyment and Satisfaction Questionnaire; QIDS-SR-16, Quick Inventory of Depressive Symptomatology Self-Report 16; SDS, Sheehan Disability Scale; Ideation, Suicidal ideation (self-reported). Metric: Change, change from baseline; Value, value at a time point. Method of aggregation: = 10, less than or equal to 10; = 12, less than or equal to 12; = 7, less than or equal to 7; = 8, less than or equal to 8; 15% increase, at least 15% increase; 25% increase, at least 25% increase; 30% decrease, at least 30% decrease; 40% decrease, at least 40% decrease; 50% decrease, at least 50% decrease; 60% decrease, at least 60% decrease; 7% increase, at least 7% increase; 70% decrease, at least 70% decrease; extremely ill, among the most extremely ill; M or VM improved, much or very much improved; MADRS = 12 and YMRS = 12, MADRS less than or equal to 12 and YMRS less than or equal to 12; MADRS = 12 and YMRS = 8, MADRS less than or equal to 12 and YMRS less than or equal to 8; Mean % Max, mean percent of the maximum; Min improved, minimally improved; Min worse, minimally worse; Normal or Borderline, normal, not at all ill or borderline ill; Normal, not ill, Normal, not at all ill; Proportion, proportion of participants who experienced the event; Remission (ND), remission (not defined); Response (ND), response (not defined); V Much improved, very much improved; V Much worse, very much worse.

categorical results). The number of meta-analyzable results could be larger than the number of treatment effects because, for multiple contrasts across trials, we expected

to find multiple results corresponding to the same treatment effect (for a single contrast in a trial, we expected to find one result corresponding to each treatment effect).

Table 2. Outcomes and treatment effects estimated across clinical trial reports

	Gabapentin-neuropathic pain (21 RCTs)			Quetiapine-bipolar depression (7 RCTs)		
	All reports	Public reports	Nonpublic reports	All reports	Public reports	Nonpublic reports
Number of reports	74	68 (92%)	6 (8%)	50	46 (92%)	4 (8%)
Number of outcomes						
Prespecified outcome domains	4	4 (100%)	4 (100%)	7	7 (100%)	7 (100%)
Unique outcomes (defined and not defined)	250	140 (56%)	187 (75%)	107	63 (59%)	66 (62%)
Unique defined outcomes	214	104 (49%)	187 (87%)	81	37 (46%)	65 (80%)
Number of treatment effects						
Nonunique treatment effects (defined and not defined) ^a	1,746	699 (40%)	1,047 (60%)	955	355 (37%)	600 (63%)
Nonunique defined treatment effects	1,122	234 (21%)	888 (79%)	739	141 (19%)	598 (81%)
Unique defined treatment effects	605	107 (18%)	553 (91%)	188	51 (27%)	159 (85%)
Number of results						
Nonunique meta-analyzable results ^a	1,364	342 (25%)	1,022 (75%)	685	113 (16%)	572 (84%)
Unique meta-analyzable results	1,230	305 (25%)	1,017 (83%)	661	109 (16%)	571 (86%)
Unique meta-analyzable results for which all elements were defined	1,032	192 (19%)	870 (84%)	630	78 (12%)	571 (91%)

A defined outcome includes five elements: domain, measure, metric, method of aggregation, and time point of assessment. A defined treatment effect includes all items required for a defined outcome, and it also includes the analysis population, method of handling missing data, and method of analysis. Unique outcomes and treatment effects were counted only once, regardless of how many times they appeared across all trials and reports. Nonunique outcomes and treatment effects were counted each time they appeared across trials and reports. We were able to count the number of unique and nonunique outcomes and treatment effects if they were defined; we were able to count only the number of nonunique outcomes and treatment effects if they were not defined. We considered an outcome or treatment effect to have occurred in most trials if it occurred in any report of at least 11 (gabapentin-neuropathic pain) or 4 (bipolar depression) trials. Public reports = journal articles, short reports (conference abstracts, letters to the editor, posters, press releases, reports in trade publications), FDA reviews, trial registrations. Nonpublic reports = Clinical Study Reports (CSRs), CSR-Synopses.

^a It was not necessary for an outcome to be defined for a result to be meta-analyzable.

3.4. Comparing reports

We counted the number of defined outcomes and the number of defined treatment effects in each report and across all reports for each RCT. For each meta-analyzable result, we recorded the number of participants included in the analysis, and we compared the number of people that would be included in the meta-analyses that included (1) the most participants possible and (2) the fewest participants possible.

We use the term “unique” when we counted an outcome, treatment effect, or result only once, regardless of how many times the outcome appeared across all RCTs and reports. For example, if two reports included the pain intensity domain measured using the McGill Pain Questionnaire and assessed as the mean difference between groups in their change in pain between baseline and the 8-week assessment, we counted that as one unique defined outcome. We use the term “nonunique” when we counted an outcome, treatment effect, or result each time it appeared across reports (e.g., in both a journal article and a conference abstract about the same trial).

We used Stata 14 (StataCorp, College Station, Texas) and R 3.3.1 (R Core Team, Vienna, Austria) to analyze the results and D3.js JavaScript Library version 3 (www.D3js.org) to draw Fig. 2.

4. Results

4.1. Results of the search

As described elsewhere [21], in 2015, we identified 21 RCTs of gabapentin for neuropathic pain (68 public and 6 nonpublic reports) and seven RCTs of quetiapine for bipolar depression (46 public and 4 nonpublic reports) meeting our eligibility criteria (Table 2).

4.2. Trials assessed many outcomes for a few outcome domains

In the reports we identified, investigators reported 4/5 (gabapentin-neuropathic pain) and 7/8 (quetiapine-bipolar depression) of our prespecified outcome domains [21]. We identified 214 and 81 unique defined outcomes for the four and seven prespecified domains that were reported, respectively; the proliferation of defined outcomes is shown in Fig. 2, the details of which may be easier to view on screen rather than in print.

The number of unique defined outcomes differed across outcome domains. For pain intensity and depression, the most common outcome domains for gabapentin-neuropathic pain and quetiapine-bipolar depression, respectively, we found 119 and 44 unique defined outcomes. For

the other three gabapentin-neuropathic pain and six quetiapine-bipolar depression outcome domains, we identified an average of 32 (range 7–76) and six (range 4–11) unique defined outcomes (Supplement A at www.jclinepi.com).

4.3. Trials used multiple methods of analysis

For gabapentin-neuropathic pain and quetiapine-bipolar depression RCTs, respectively, we identified four and seven unique analysis populations and five and three unique methods of handling missing data. For both topics, we identified adjusted and unadjusted analyses.

Differences in analysis populations and methods of handling missing data affected the number of people included in each analysis. Meta-analyses including the same RCTs could have included 2,424 to 3,239 participants for the outcome domain “pain intensity” and 840 to 1,721 participants for the outcome domain “depression” [21], differences of 34% and 105%.

4.4. Defined outcomes were analyzed using multiple methods

For the prespecified outcome domains, the total number of treatment effects that were assessed increased with the

addition of each analysis population, method of handling missing data, and method of adjustment (Table 1). In total, 21 gabapentin-neuropathic pain and 7 quetiapine-bipolar depression RCTs evaluated 605 and 188 unique defined treatment effects, respectively, for the seven and four prespecified domains that were assessed in these RCTs (Fig. 3). There was more variation in the assessment and reporting of pain intensity and depression compared with other outcome domains (Supplement B at www.jclinepi.com).

4.5. Many treatment effects were not described in public reports

Including all RCTs, public reports described 699/1,746 (40% gabapentin-neuropathic pain) and 355/955 (37% quetiapine-bipolar depression) of the nonunique treatment effects available from all reports (Table 2).

For the RCTs in which we had nonpublic reports (six gabapentin-neuropathic pain and four quetiapine-bipolar depression RCTs), we found that public reports described 510/1,557 (33%) and 305/905 (34%) of the nonunique treatment effects available from all reports of these trials. One gabapentin-neuropathic pain trial had only a nonpublic report (i.e., there was no associated journal article or other

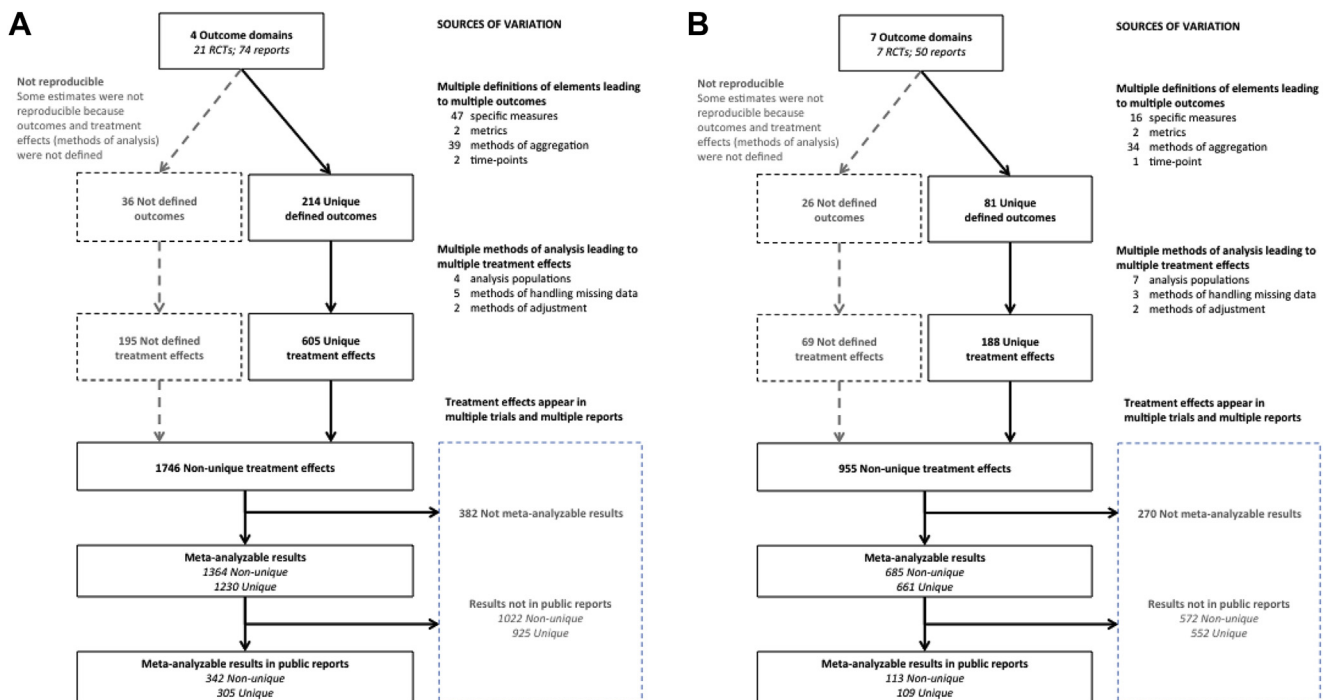


Fig. 3. Usable and reproducible results in public reports are a small subset of all clinical trial results. The top row represents the number of prespecified outcome domains. Subsequent rows show the number of defined outcomes and treatment effects, and annotations to the right of the flowchart identify sources of variation in outcome definitions and treatment effects. The number of nonreproducible outcomes and treatment effects likely overestimates the true number of outcomes and treatment effects that were assessed; unless all elements were defined, we could not tell if two treatment effects were the same or different, and some of these outcomes and treatment effects may have been completely defined in other sources. Results could have been associated with outcomes that were defined or not defined. Of all results we identified, some were not meta-analyzable, and some meta-analyzable results were not included in public reports. The final row represents the number of meta-analyzable results available from public reports. (A) Gabapentin for neuropathic pain. (B) Quetiapine for bipolar depression.

public report) and the nonpublic report included 305 treatment effects.

Some metrics and methods of aggregation described in public reports were not described in nonpublic reports. For example, one gabapentin-neuropathic pain report [30] described categories for change in pain intensity (e.g., ≥ 4 point change, ≥ 3 point change) that were not found in the corresponding nonpublic reports (Supplement C at www.jclinepi.com). All measures and time points in public reports were included in nonpublic sources.

4.6. Reports included insufficient data for meta-analysis

Of all nonunique results in public and nonpublic reports, 1,364/1,746 (78%) and 685/955 (72%) were meta-analyzable (Table 2).

Of the treatment effects reported publicly, we found a meta-analyzable result for 342/699 (49% gabapentin-neuropathic pain) and 113/355 (32% quetiapine-bipolar depression; Table 2). The most common reasons that results were not meta-analyzable were the failure to report the denominator for dichotomous outcomes and the failure to report a measure of precision for continuous outcomes.

4.7. Some meta-analyzable results were not reproducible

Reports sometimes included meta-analyzable data (e.g., mean, standard error, number of participants) but did not define all elements of the outcome or treatment effect, making it impossible for another investigator to reproduce the results. In particular, methods of analysis were better defined in nonpublic reports compared with public reports. At least one element was not defined for 294/1,364 (22%) (gabapentin-neuropathic pain) and 33/685 (5%) (quetiapine-bipolar depression) nonunique meta-analyzable results (Supplement D at www.jclinepi.com). Nonpublic reports typically defined these elements completely, and conference abstracts were the type of report least likely to define all elements of included outcomes and treatment effects (Supplement E at www.jclinepi.com).

4.8. Outcomes and treatment effects differed across trials

Because of different outcome definitions and because of incomplete reporting, no meta-analysis could have included all of the eligible RCTs for either gabapentin-neuropathic pain or quetiapine-bipolar depression [21]. Indeed, few outcomes appeared in most trials; 2/214 (1%) and 12/81 (15%), respectively, of the unique defined outcomes were assessed in a majority of RCTs (i.e., at least 11/21 gabapentin-neuropathic pain RCTs and at least 4/7 quetiapine-bipolar depression RCTs). Many outcomes appeared in only one trial; 116/214 (54%) and 23/81 (28%), respectively, were assessed in only one RCT. This was similar to what we observed for treatment effects, with 0/

605 (0%) and 8/188 (4%), respectively, unique defined treatment effects assessed in a majority of RCTs while 508/605 (84%) and 64/188 (34%) were assessed in only one RCT (Table 3).

5. Discussion

We examined two commonly prescribed drugs recommended in clinical practice guidelines, gabapentin for neuropathic pain [31] and quetiapine for bipolar depression [32]. These are among a small number of drugs and indications for which nonpublic sources are available; our findings were very similar for both cases and they were consistent with previous studies, suggesting that the areas of concern we identified are applicable to the wider situation for clinical trials (Box 2). We found multiple reports of RCTs, some public and some nonpublic [21]. Within these reports, we found multiple outcomes reported for each trial and across all trials (Table 3). The variety of outcome definitions and methods of analysis led to the large number of results reported for these RCTs. Many outcomes were not defined, even for results that could be included in meta-analyses (Table 2). Because we restricted our study to prespecified outcome domains, we recorded only one time point within each prespecified window, and we did not count, subgroup populations as separate analysis populations, our findings are likely to understate the total number of results for these trials. The fact that many results exist for every research question creates important challenges for interpreting the findings of RCTs and for evidence synthesis.

Although we found an excessive number of outcomes and treatment effects in these reports, much important information was not publicly available [33]. Fewer than half of the unique outcomes appeared in public reports (Table 2). Even when information was publicly available, many results were not meta-analyzable, and thus the information could not inform practice [34]. Furthermore, some meta-analyzable results were nonreproducible [35] because there was insufficient information to determine which outcome had been assessed or which methods of analyses had been used. For example, it was often unclear how the authors handled missing data. Systematic reviewers who use outcomes that are not defined might naively compare dissimilar information and reach incorrect conclusions. This could be a particular problem for meta-analyses that compare different measures on a uniform scale (e.g., risk ratio, SMD) [7]. For this reason, statistical methods that allow the combination of results across trials with disparate outcomes might also disguise important differences among trials.

In addition to the problems we identified within trials, we found that differences across trials make it very difficult to compare RCTs (e.g., in systematic reviews and clinical guidelines), and these differences pose important obstacles to knowledge translation. Although we specified only a few outcome domains, RCTs assessed hundreds of different treatment effects within them [36]. Previous authors have

Table 3. Occurrence of treatment effects across clinical trial reports

	Gabapentin-neuropathic pain (21 RCTs)	Quetiapine-bipolar depression (7 RCTs)
Outcomes		
Mean unique outcomes per trial (range)	26 (0–95)	27 (6–65)
Unique outcomes occurring in all trials	0	1
Unique outcomes occurring in exactly one trial	116	23
Unique outcomes occurring in the majority of trials ^a	2	12
Treatment effects		
Mean unique treatment effects per trial (range)	38 (0–305)	48 (0–148)
Unique treatment effects occurring in all trials	0	0
Unique treatment effects occurring in exactly one trial	508	64
Unique treatment effects occurring in the majority of trials ^a	0	8
Results		
Mean unique meta-analyzable results per trial (range)	59 (0–308)	94 (0–296)
Number of participants who could be included in the analysis by selecting one result from each trial (range) ^b	2,424–3,239	840–1,721

^a We considered an outcome or treatment effect to have appeared in most trials if it was present in any report of at least 11 of the 21 gabapentin-neuropathic pain RCTs or at least 4 of 7 quetiapine-bipolar depression RCTs.

^b For gabapentin-neuropathic pain, we used the most common domain, “pain intensity.” For quetiapine-bipolar depression, we used the most common domain, “depression.”

argued that core outcome sets, which list the minimum outcomes to assess in research, could improve the comparability of trials [37,38] and improve the synthesis of trials for knowledge translation [39]. Although we agree, our results also show that efforts to develop core outcome sets might have little impact unless they define outcomes completely. That is, eligible trials assessed many of the same domains, yet no defined outcome was reported in all trials of gabapentin-neuropathic pain and only one defined outcome appeared in all trials of quetiapine-bipolar depression (Table 3). Worse, few outcomes for gabapentin-neuropathic pain and quetiapine-bipolar depression appeared in most trials; consequently, most meta-analyses could have included only a small proportion of the eligible RCTs and only a small proportion of the outcomes they assessed (Table 3). Of all outcomes, almost half were reported in exactly one trial; these outcomes could not be combined with the same outcome

from any other trial except by using statistical procedures that obscure differences in measurement (e.g., SMD). Considering differences in analyses, the situation is even worse; 1% of unique treatment effects appeared in most trials, whereas a majority appeared in exactly one trial. In these ways, our findings indicate that the state of evidence is the opposite of what would be best for knowledge translation and for patient care; that is, it would be desirable for all trials to include a few common outcomes that are defined and analyzed in the same way.

Understanding the plethora of outcomes in RCTs could identify ways to improve trials and evidence synthesis. On one hand, it is efficient to collect multiple outcomes, and it is good practice for trialists to conduct sensitivity analyses to evaluate whether their findings are robust under different assumptions [40,41]. On the other hand, multiple outcomes and analyses lead to different results and, thus, allow trialists [15,16] and systematic reviewers to cherry-pick [17,19,21,38]. We are concerned that the sources of variation we identified are not always described in protocols and registrations for trials [42,43] or systematic reviews. For example, protocols for systematic reviews usually specify the outcome domains and the time points of interest [24]; however, few define their methods for handling multiple measures or multiple analyses [7,13,38]. For these reasons, independent investigators have been unable to reproduce the data selected for systematic reviews or to replicate the meta-analyses included in reviews [44]. At a minimum, systematic reviewers should expect clinical trials to include many definitions of outcomes within the domains of interest, and reviewers should prespecify plans for handling many different outcomes and results across multiple reports of eligible trials.

Our study adds to previous research by using both public and nonpublic reports [7,13]. Because there is strong evidence that trialists purposively select outcomes to include

Box 2. Key findings and implications of this study

Key findings from our study

Many reports are available for individual RCTs
Some RCT reports are public, but many are nonpublic
Multiple outcomes are described across RCT, even within 11 prespecified outcome domains
Some outcomes are defined in RCT reports, and some are not defined
Many treatment effects are described in individual trials and across all trials
Many results are described in individual trials
Not all results are meta-analyzable
Nonpublic reports include more information than public reports about outcomes, treatment effects, and results

Implications of our findings

Many reports do not include enough information to reproduce RCT results
Clinical decision making may not be well informed
Clinical decision making may not be accurately informed
Much of the available research may be wasted

Abbreviation: RCT, randomized clinical trial.

in public reports [15,16], it is not surprising that previous studies limited to public reports suggest that systematic reviews are not affected by which results are selected for analysis [38]. By comparison, our study suggests that systematic reviewers using nonpublic reports have considerable opportunities for cherry-picking and that meta-analyses that include nonpublic sources may be affected by reviewers' selection of outcomes [21].

In response to growing mandates for “open science” [45–47], trialists may begin to share previously nonpublic reports and to provide access to IPD. Consequently, it is increasingly important for systematic reviewers to include plans in their protocols for handling the sources of variation we identified.

Our study focused on two research questions, and eligible RCTs were reported between 1997 and 2013 (gabapentin-neuropathic pain RCTs) and 2003 and 2014 (quetiapine-bipolar depression RCTs). Although the included trials might not reflect current practice, we are not aware of any evidence that suggests the number of outcomes and analyses have decreased over time. Indeed, the opposite might be true. We had access to nonpublic reports for only one-third of RCTs, so the total number of outcomes and treatment effects in these RCTs might have been greater than our findings suggest.

In the short term, adherence to existing guidance could improve current practice. Journals should adopt standards that promote transparency in all the research they publish [48]. Trialists should register their studies [49–51] and consult the EQUATOR Network [52] for relevant reporting guidelines for protocols [28] and research reports [26]. Systematic reviewers should use reporting guidelines when publishing protocols [53] and final reports using aggregate data [54] or IPD [55]. Our results demonstrate that such efforts could reduce opportunities for cherry-picking. When used systematically, our findings support the theory that nonpublic sources may improve the accuracy and quality of evidence syntheses [56]. However, our results also suggest that nonpublic data could increase burden on reviewers and introduce new opportunities for cherry-picking. Additional guidance for using multiple outcomes and analyses from multiple reports of clinical trials is needed.

Acknowledgments

The authors used the open access database, Systematic Review Data Repository (SRDR), and are grateful to Jens Jap, Bryant Smith, and Joseph Lau for making this possible without charge and for the support they provided. The authors are grateful to Élise Diard (Centre de Recherche Épidémiologie et Statistique Sorbonne Paris Cité) for contributing substantially to the figures.

Authors' contributions: Study conception and design: The study design was first described in the application to

the Patient-Centered Outcomes Research Institute (PCORI) in 2013 (Kay Dickersin, principal investigator), to which Peter Doshi, Tianjing Li, and Swaroop Vedula contributed. E.M.-W. drafted the protocol with contributions from other authors. Acquisition of data: Lori Rosman and Claire Taylor designed and ran the electronic searches. Terrie Cowley, Kay Dickersin, Nicole Fusco, Gillian Gresham, Jennifer Haythornthwaite, James Heyward, Tianjing Li, Diana Lock, Evan Mayo-Wilson, Jennifer Payne, and Elizabeth Tolbert contributed to drafting and finalizing the data extraction forms. Kay Dickersin, Nicole Fusco, Gillian Gresham, James Heyward, Susan Hutfless, Tianjing Li, Evan Mayo-Wilson, and Swaroop Vedula screened studies for inclusion. Lorenzo Bertizzolo, Joseph K. Canner, Jeffrey Ehmsen, Nicole Fusco, Gillian Gresham, James Heyward, Diana Lock, Evan Mayo-Wilson, and Catalina Suarez-Cuervo extracted the data. Analysis and interpretation of data: Joseph K. Canner, Nicole Fusco, Hwanhee Hong, and Evan Mayo-Wilson managed the data. Joseph K. Canner, Nicole Fusco, and Hwanhee Hong analyzed the data. Joseph K. Canner, Kay Dickersin, Nicole Fusco, Hwanhee Hong, Tianjing Li, and Evan Mayo-Wilson contributed to the interpretation and data presentation. Drafting of article: Evan Mayo-Wilson wrote the first draft, with Kay Dickersin, Nicole Fusco, Tianjing Li, and Evan Mayo-Wilson providing subsequent revisions. Joseph K. Canner drew the figures. Critical revision: All authors reviewed, provided critical revisions, and approved the article for publication.

E.M.-W. is the guarantor. All authors, external and internal, had full access to all of the data (including statistical reports and tables) in the study and can take responsibility for the integrity of the data and the accuracy of the data analysis.

Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jclinepi.2017.05.007>.

References

- [1] Zarin DA, Tse T, Williams RJ, Califf RM, Ide NC. The Clinical-Trials.gov results database—update and key issues. *N Engl J Med* 2011;364:852–60.
- [2] Davey J, Turner RM, Clarke MJ, Higgins JP. Characteristics of meta-analyses and their component studies in the Cochrane Database of Systematic Reviews: a cross-sectional, descriptive analysis. *BMC Med Res Methodol* 2011;11:160.
- [3] Kirkham JJ, Dwan KM, Altman DG, Gamble C, Dodd S, Smyth R, et al. The impact of outcome reporting bias in randomised controlled trials on a cohort of systematic reviews. *BMJ* 2010; 340:c365.
- [4] Dwan K, Altman DG, Cresswell L, Blundell M, Gamble CL, Williamson PR. Comparison of protocols and registry entries to published reports for randomised controlled trials. *Cochrane Database Syst Rev* 2011;MR000031.

- [5] Furukawa TA, Watanabe N, Omori IM, Montori VM, Guyatt GH. Association between unreported outcomes and effect size estimates in Cochrane meta-analyses. *Jama* 2007;297:468–70.
- [6] Moayyedi P, Deeks J, Talley NJ, Delaney B, Forman D. An update of the Cochrane systematic review of *Helicobacter pylori* eradication therapy in nonulcer dyspepsia: resolving the discrepancy between systematic reviews. *Am J Gastroenterol* 2003;98(12):2621–6.
- [7] Tendal B, Nuesch E, Higgins JP, Juni P, Gotzsche PC. Multiplicity of data in trial reports and the reliability of meta-analyses: empirical study. *Bmj* 2011;343:d4829.
- [8] Schulz KF, Grimes DA. Multiplicity in randomised trials I: endpoints and treatments. *Lancet* 2005;365:1591–5.
- [9] Gyawali B, Prasad V. Same data; different interpretations. *J Clin Oncol* 2016;. <http://dx.doi.org/10.1200/JCO.2016.68.2021>.
- [10] Wood AM, White IR, Thompson SG. Are missing outcome data adequately handled? A review of published randomized controlled trials in major medical journals. *Clin Trials* 2004;1:368–76.
- [11] Chan AW, Hrobjartsson A, Jorgensen KJ, Gotzsche PC, Altman DG. Discrepancies in sample size calculations and data analyses reported in randomised trials: comparison of publications with protocols. *Bmj* 2008;337:a2299.
- [12] Vedula SS, Li T, Dickersin K. Differences in reporting of analyses in internal company documents versus published trial reports: comparisons in industry-sponsored trials in off-label uses of gabapentin. *Plos Med* 2013;10(1):e1001378.
- [13] Page MJ, McKenzie JE, Chau M, Green SE, Forbes A. Methods to select results to include in meta-analyses deserve more consideration in systematic reviews. *J Clin Epidemiol* 2015;68:1282–91.
- [14] Dwan K, Altman DG, Clarke M, Gamble C, Higgins JP, Sterne JA, et al. Evidence for the selective reporting of analyses and discrepancies in clinical trials: a systematic review of cohort studies of clinical trials. *Plos Med* 2014;11(6):e1001666.
- [15] Dwan K, Gamble C, Williamson PR, Kirkham JJ. Reporting Bias Group. Systematic review of the empirical evidence of study publication bias and outcome reporting bias - an updated review. *PLoS One* 2013;8:e66844.
- [16] Chan AW, Hrobjartsson A, Haahr MT, Gotzsche PC, Altman DG. Empirical evidence for selective reporting of outcomes in randomized trials: comparison of protocols to published articles. *JAMA* 2004;291:2457–65.
- [17] Chalmers TC, Frank CS, Reitman D. Minimizing the three stages of publication bias. *JAMA* 1990;263:1392–5.
- [18] Page MJ, McKenzie JE, Forbes A. Many scenarios exist for selective inclusion and reporting of results in randomized trials and systematic reviews. *J Clin Epidemiol* 2013;66:524–37.
- [19] Page MJ, McKenzie JE, Kirkham J, Dwan K, Kramer S, Green S, et al. Bias due to selective inclusion and reporting of outcomes and analyses in systematic reviews of randomised trials of health-care interventions. *Cochrane Database Syst Rev* 2014;MR000035.
- [20] Mayo-Wilson E, Hutfless S, Li T, Gresham G, Fusco N, Ehmsen J, et al. Integrating multiple data sources (MUDS) for meta-analysis to improve patient-centered outcomes research: a protocol for a systematic review. *Syst Rev* 2015;4:143.
- [21] Mayo-Wilson E, Li T, Fusco N, Bertizzolo L, Canner J, Cowley T et al. Disagreements across multiple reports of clinical trials: implications for interpretation of trial quality and results. Under Rev.
- [22] Meinert CL. *Clinical Trials Dictionary: Terminology and Usage Recommendations*. 2nd ed. Hoboken, NJ: John Wiley & Sons; 2012.
- [23] Li T, Vedula SS, Hadar N, Parkin C, Lau J, Dickersin K. Innovations in data collection, management, and archiving for systematic reviews. *Ann Intern Med* 2015;162:287–94.
- [24] Saldanha IJ, Dickersin K, Wang X, Li T. Outcomes in Cochrane systematic reviews addressing four common eye conditions: an evaluation of completeness and comparability. *PLoS One* 2014;9:e109400.
- [25] Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *BMJ* 2015;349:g7647.
- [26] Schulz KF, Altman DG, Moher D, Group C. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *Plos Med* 2010;7(3):e1000251.
- [27] Chan AW, Tetzlaff JM, Altman DG, Dickersin K, Moher D. SPIRIT 2013: new guidance for content of clinical trial protocols. *Lancet* 2013;381:91–2.
- [28] Chan AW, Tetzlaff JM, Altman DG, Laupacis A, Gotzsche PC, Krleza-Jeric K, et al. SPIRIT 2013 statement: defining standard protocol items for clinical trials. *Ann Intern Med* 2013;158:200–7.
- [29] Patel CJ, Burford B, Ioannidis JP. Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. *J Clin Epidemiol* 2015;68:1046–58.
- [30] Backonja M. Gabapentin for painful diabetic neuropathy (in reply). *JAMA* 1999;282:133–4.
- [31] Kendall T, Morriss R, Mayo-Wilson E, Marcus E. Guideline development group of the National Institute for Health and Care Excellence. assessment and management of bipolar disorder: summary of updated NICE guidance. *BMJ* 2014;349:g5673.
- [32] Dworkin RH, O'Connor AB, Audette J, Baron R, Gourlay GK, Haanpaa ML, et al. Recommendations for the pharmacological management of neuropathic pain: an overview and literature update. *Mayo Clin Proc* 2010;85(3 Suppl):S3–14.
- [33] Chan AW, Song F, Vickers A, Jefferson T, Dickersin K, Gotzsche PC, et al. Increasing value and reducing waste: addressing inaccessible research. *Lancet* 2014;383:257–66.
- [34] Glasziou P, Altman DG, Bossuyt P, Boutron I, Clarke M, Julious S, et al. Reducing waste from incomplete or unusable reports of biomedical research. *Lancet* 2014;383:267–76.
- [35] Goodman S, Fanelli D, Ioannidis JP. What does research reproducibility mean? *Sci Trans Med* 2016;8(341):341ps12.
- [36] Kaufman J, Ryan R, Bosch-Capblanch X, Cartier Y, Cliff J, Glenton C, et al. Outcomes mapping study for childhood vaccination communication: too few concepts were measured in too many ways. *J Clin Epidemiol* 2016;72:33–44.
- [37] Williamson P, Clarke M. The COMET (Core outcome measures in effectiveness trials) initiative: its Role in improving Cochrane reviews. *Cochrane Database Syst Rev* 2012;ED000041.
- [38] Page MJ, Forbes A, Chau M, Green SE, McKenzie JE. Investigation of bias in meta-analyses due to selective inclusion of trial effect estimates: empirical study. *BMJ Open* 2016;6(4):e011863.
- [39] Saldanha IJ, Li T, Yang C, Owczarzak J, Williamson PR, Dickersin K. Clinical trials and systematic reviews addressing similar interventions for the same condition do not consider similar outcomes to be important: a case study in HIV/AIDS. *J Clin Epidemiol* 2017;84:85–94.
- [40] Altman DG. Missing outcomes in randomized trials: addressing the dilemma. *Open Med* 2009;3(2):e51–3.
- [41] Li T, Hutfless S, Scharfstein D, Daniels M, Hogan J, Little R, et al. Standards should be applied in the prevention and handling of missing data for patient-centered outcomes research: a systematic review and expert consensus. *J Clin Epidemiol* 2014;67:15–32.
- [42] Mathieu S, Boutron I, Moher D, Altman DG, Ravaud P. Comparison of registered and published primary outcomes in randomized controlled trials. *JAMA* 2009;302:977–84.
- [43] Perlmutter A, Tran VT, Dechartres A, Ravaud P. Comparison of primary outcomes in protocols, public clinical-trial registries and publications: the example of oncology trials. *Ann Oncol* 2017;28:688–95.
- [44] Tendal B, Higgins JP, Juni P, Hrobjartsson A, Trelle S, Nuesch E, et al. Disagreements in meta-analyses using outcomes measured on continuous or rating scales: observer agreement study. *BMJ* 2009;339:b3128.
- [45] Taichman DB, Backus J, Baethge C, Bauchner H, de Leeuw PW, Drazen JM, et al. Sharing clinical trial data: a proposal from the international committee of medical journal editors. *PLoS Med* 2016;13(1):e1001950.

- [46] Moorthy VS, Karam G, Vannice KS, Kieny MP. Rationale for WHO's new position calling for prompt reporting and public disclosure of interventional clinical trial results. *PLoS Med* 2015;12(4):e1001819.
- [47] Doshi P, Dickersin K, Healy D, Vedula SS, Jefferson T. Restoring invisible and abandoned trials: a call for people to publish the findings. *BMJ* 2013;346:f2865.
- [48] Nosek BA, Alter G, Banks GC, Borsboom D, Bowman SD, Breckler SJ, et al. Scientific standards. Promoting an open research culture. *Science* 2015;348:1422–5.
- [49] Weber WE, Merino JG, Loder E. Trial registration 10 years on. *BMJ* 2015;351:h3572.
- [50] Zarin DA, Tse T, Williams RJ, Carr S. Trial reporting in ClinicalTrials.gov—the final rule. *N Engl J Med* 2016;375:1998–2004.
- [51] Hudson KL, Lauer MS, Collins FS. Toward a new era of trust and transparency in clinical trials. *JAMA* 2016;316:1353–4.
- [52] Simera I, Altman DG, Moher D, Schulz KF, Hoey J. Guidelines for reporting health research: the EQUATOR network's survey of guideline authors. *PLoS Med* 2008;5(6):e139.
- [53] Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4:1.
- [54] Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Plos Med* 2009;6(7):e1000097.
- [55] Stewart LA, Clarke M, Rovers M, Riley RD, Simmonds M, Stewart G, et al. Preferred reporting items for systematic review and meta-analyses of individual participant data: the PRISMA-IPD Statement. *JAMA* 2015;313:1657–65.
- [56] Doshi P, Jefferson T, Del Mar C. The imperative to share clinical study reports: recommendations from the Tamiflu experience. *PLoS Med* 2012;9(4):e1001201.